

SOME SOLUTIONS TO THE PLANETARY GEOSTROPHIC EQUATIONS

TWNH

Peterson and Callies (in review) (see also *Peterson and Callies* 2023) consider how bottom-intensified vertical mixing induces a basin-scale circulation in various simple non-flat-bottomed basins. They write equations for a bottom Ekman layer and couple them to an equation for the geostrophic contour vorticity balance, in the parlance of *Waldman and Giordani* (2023), which is a type of barotropic vorticity equation.

1. GOVERNING EQUATIONS

The flow satisfies the (non-dimensional) planetary-geostrophic and hydrostatic equations:

$$\begin{aligned} (1) \quad & -fv = -\frac{\partial p}{\partial x} + \epsilon^2 \frac{\partial}{\partial z} \left(\nu \frac{\partial u}{\partial z} \right), \\ (2) \quad & fu = -\frac{\partial p}{\partial y} + \epsilon^2 \frac{\partial}{\partial z} \left(\nu \frac{\partial v}{\partial z} \right), \\ (3) \quad & \frac{\partial p}{\partial z} = b \end{aligned}$$

(see Eqs. (9)–(11) of *Peterson and Callies* in review and Eq. (3.26) of *Klinger and Haine* 2019). The flow is non-divergent:

$$(4) \quad \nabla \cdot \mathbf{u} = 0,$$

and the buoyancy field satisfies the advection-diffusion equation

$$(5) \quad \mu\rho \left(\frac{\partial b}{\partial t} + \mathbf{u} \cdot \nabla b \right) = \epsilon^2 \frac{\partial}{\partial z} \left(\kappa \frac{\partial b}{\partial z} \right).$$

(see Eqs. (11)–(13) of *Peterson and Callies* in review). The boundary conditions on buoyancy are:

$$(6) \quad \text{Surface } z = 0 : b = 0$$

$$(7) \quad \text{Bottom } z = -H : \frac{\partial b}{\partial z} = 0,$$

and the initial condition is $b = z$.

The diffusivity and viscosity are given by

$$(8) \quad \kappa(z) = \kappa_0 + \exp(-\alpha(z + H)),$$

$$(9) \quad \nu = \text{constant}$$

Peterson and Callies (in review) Eq. (17).

2. VERTICALLY-INTEGRATED FLOW

The vertically-integrated flow (U, V) is given by:

$$(10) \quad \begin{aligned} U &\equiv \int_{-H}^0 u \, dz = - \int_{-H}^0 z \partial_z u \, dz, \\ V &\equiv \int_{-H}^0 v \, dz = - \int_{-H}^0 z \partial_z v \, dz, \end{aligned}$$

exploiting the bottom boundary condition on (u, v) . The equations for (U, V) come from vertically integrating the momentum equations (1), (2):

$$(11) \quad -fV = - \int_{-H}^0 \frac{\partial p}{\partial x} \, dz + \epsilon^2 \int_{-H}^0 \frac{\partial}{\partial z} \left(\nu \frac{\partial u}{\partial z} \right) \, dz$$

$$(12) \quad fU = - \int_{-H}^0 \frac{\partial p}{\partial y} \, dz + \epsilon^2 \int_{-H}^0 \frac{\partial}{\partial z} \left(\nu \frac{\partial v}{\partial z} \right) \, dz.$$

The depth-integrated pressure gradient in the x direction is:

$$(13) \quad \int_{-H}^0 \frac{\partial p}{\partial x} \, dz = - \frac{\partial H}{\partial x} p(x, z = -H) + \frac{\partial}{\partial x} \int_{-H}^0 p \, dz,$$

$$(14) \quad = -p_b \frac{\partial H}{\partial x} + \frac{\partial}{\partial x} \left[p_b H - \int_{-H}^0 z \frac{\partial p}{\partial z} \right],$$

$$(15) \quad = H \frac{\partial p_b}{\partial x} - \frac{\partial}{\partial x} \int_{-H}^0 z b \, dz,$$

using the Liebniz integration formula¹, where $p_b = p(x, -H(x))$ is the bottom pressure, and integration by parts². The viscous term in the x direction is:

$$(16) \quad \int_{-H}^0 \frac{\partial}{\partial z} \left(\nu \frac{\partial u}{\partial z} \right) \, dz = \left[\nu \frac{\partial u}{\partial z} \right]_{-H}^0 = \frac{\tau^x}{\epsilon^2} - \nu \frac{\partial u}{\partial z} \Big|_{z=-H},$$

which uses the boundary condition (31).

Therefore (11) and (12) lead to:

$$(17) \quad -fV = -H \frac{\partial p_b}{\partial x} - \frac{\partial \chi}{\partial x} - \tau_b^x + \tau^x,$$

$$(18) \quad fU = -H \frac{\partial p_b}{\partial y} - \frac{\partial \chi}{\partial y} - \tau_b^y + \tau^y,$$

¹The Liebniz integration formula reads:

$$\partial_x \left(\int_{a(x)}^{b(x)} f(x, y) \, dy \right) = \partial_x b f(x, b(x)) - \partial_x a f(x, a(x)) + \int_{a(x)}^{b(x)} \partial_x f(x, y) \, dy.$$

²Integration by parts gives:

$$\begin{aligned} \int_{-H}^0 z \frac{\partial p}{\partial z} \, dz &= [zp]_{-H}^0 - \int_{-H}^0 p \, dz, \\ &= p_b H - \int_{-H}^0 p \, dz. \end{aligned}$$

where

$$(19) \quad \chi = - \int_{-H}^0 z b \, dz$$

is the baroclinic potential energy, and $\epsilon^2 \nu \partial_z(u, v)|_{z=-H} \equiv (\tau_b^x, \tau_b^y) = \tau_b$ is the bottom stress on the fluid.

Next, divide by water depth H :

$$(20) \quad -\frac{fV}{H} = -\frac{\partial p_b}{\partial x} - \frac{1}{H} \frac{\partial \chi}{\partial x} + \frac{\tau^x - \tau_b^x}{H}$$

$$(21) \quad \frac{fU}{H} = -\frac{\partial p_b}{\partial y} - \frac{1}{H} \frac{\partial \chi}{\partial y} + \frac{\tau^y - \tau_b^y}{H},$$

and take the curl³

$$(23) \quad \begin{aligned} \frac{\partial}{\partial y} \left\{ -\frac{fV}{H} \right\} - \frac{\partial}{\partial x} \left\{ \frac{fU}{H} \right\} &= \frac{\partial}{\partial y} \left\{ -\frac{\partial p_b}{\partial x} - \frac{1}{H} \frac{\partial \chi}{\partial x} + \frac{\tau^x - \tau_b^x}{H} \right\} \\ &\quad - \frac{\partial}{\partial x} \left\{ -\frac{\partial p_b}{\partial y} - \frac{1}{H} \frac{\partial \chi}{\partial y} + \frac{\tau^y - \tau_b^y}{H} \right\}, \end{aligned}$$

$$(24) \quad \Rightarrow \frac{\partial}{\partial y} \left\{ -\frac{f}{H} \frac{\partial \Psi}{\partial x} \right\} + \frac{\partial}{\partial x} \left\{ \frac{f}{H} \frac{\partial \Psi}{\partial y} \right\} = -\frac{\partial}{\partial y} \left\{ \frac{1}{H} \frac{\partial \chi}{\partial x} \right\} + \frac{\partial}{\partial x} \left\{ \frac{1}{H} \frac{\partial \chi}{\partial y} \right\} - \hat{\mathbf{z}} \cdot \left(\nabla \times \frac{\tau - \tau_b}{H} \right),$$

$$(25) \quad \Rightarrow -\frac{\partial}{\partial y} \left(\frac{f}{H} \right) \frac{\partial \Psi}{\partial x} + \frac{\partial}{\partial x} \left(\frac{f}{H} \right) \frac{\partial \Psi}{\partial y} = -\frac{\partial}{\partial y} \left(\frac{1}{H} \right) \frac{\partial \chi}{\partial x} + \frac{\partial}{\partial x} \left(\frac{1}{H} \right) \frac{\partial \chi}{\partial y} - \hat{\mathbf{z}} \cdot \left(\nabla \times \frac{\tau - \tau_b}{H} \right),$$

$$(25) \quad \Rightarrow J \left(\frac{f}{H}, \Psi \right) = J \left(\frac{1}{H}, \chi \right) - \hat{\mathbf{z}} \cdot \left(\nabla \times \frac{\tau - \tau_b}{H} \right),$$

for streamfunction of the depth-integrated horizontal flow Ψ where

$$(26) \quad \begin{aligned} V &= \frac{\partial \Psi}{\partial x}, \\ U &= -\frac{\partial \Psi}{\partial y}, \end{aligned}$$

and Jacobian $J(A, B) = \partial_x A \partial_y B - \partial_y A \partial_x B$. Notice how this recipe (vertically integrate, divide by H , then take the curl) eliminates the bottom pressure. The final result (25) is the same as Eq. (20) of *Peterson and Callies* (in review). It is called (a simplified form of) the **geostrophic contour vorticity balance** (GCVB) by *Waldman and Giordani* (2023), although *Peterson and Callies* (in review) call it the “barotropic vorticity equation”. It is not the same equation as used by *Nøst and Isachsen* (2003) (which *Waldman and Giordani* (2023) call the “transport divergence vorticity balance”).

Notice how the GCVB (25) is an equation for the streamfunction of the depth-integrated flow Ψ given the buoyancy field (and hence the JEBAR term) and the surface and bottom stresses. The GCVB shows how depth-integrated flow Ψ across f/H contours is driven by (i) the joint effect

³Recall that

$$(22) \quad \hat{\mathbf{z}} \cdot (\nabla \times \mathbf{U}) = \partial_x V - \partial_y U.$$

of baroclinicity and relief (JEBAR), (ii) curl of wind stress divided by H , (iii) curl of the bottom stress divided by H .

3. SOLUTION TO FRICTIONAL GEOSTROPHIC EQUATIONS IN STEADY STATE

Start from (1)–(2) and write the pressure as

$$(27) \quad p(x, y, z) = p_s(x, y) + \int_z^0 b(x, y, z') dz',$$

$$(28) \quad \implies \partial_x p(x, y, z) = \partial_x p_s(x, y) + \int_z^0 \partial_x b(x, y, z') dz'$$

$$(29) \quad \implies p_b(x, y) \equiv p(x, y, z = -H(x, y)) = p_s(x, y) + \int_{-H(x, y)}^0 b(x, y, z') dz',$$

where $p_s(x, y)$ is the (presently unknown) surface pressure (and assuming that $p_s \ll p(-H)$). Solve these frictional geostrophic equations using a Green's function, defined so that $\mathbf{u} = u + iv$, thus

$$(30) \quad \epsilon^2 \frac{\partial}{\partial z} \left(\nu \frac{\partial \mathbf{u}}{\partial z} \right) - i f \mathbf{u} = \mathbf{f},$$

$$(31) \quad \text{Surface } z = 0 : \epsilon^2 \nu \partial_z \mathbf{u} = \tau^x + i \tau^y,$$

$$(32) \quad \text{Bottom } z = -H : \mathbf{u} = 0,$$

where $\mathbf{f} = \frac{\partial p}{\partial x} + i \frac{\partial p}{\partial y}$ is the pressure gradient from (28).

The corresponding Green's function satisfies:

$$(33) \quad \epsilon^2 \frac{\partial}{\partial z} \left(\nu \frac{\partial G_{\mathbf{u}}}{\partial z} \right) - i f G_{\mathbf{u}} = \delta(z - \xi),$$

$$(34) \quad \text{Surface } z = 0 : \epsilon^2 \nu \partial_z G_{\mathbf{u}} = 0,$$

$$(35) \quad \text{Bottom } z = -H : G_{\mathbf{u}} = 0.$$

We find that the solution for constant viscosity $\nu = \nu_0$ is:

$$(36) \quad G_{\mathbf{u}}(z; \xi) = \frac{i^{3/2} e^{-\sqrt{i}\varphi(z+\xi)}}{2\nu_0 \epsilon^2 \varphi \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)} \begin{cases} -e^{2\sqrt{i}\varphi \xi} + e^{2\sqrt{i}\varphi(z+H(x))} + e^{2\sqrt{i}\varphi(z+\xi+H(x))} - 1 & \text{for } z \leq \xi, \\ -e^{2\sqrt{i}\varphi z} + e^{2\sqrt{i}\varphi(\xi+H(x))} + e^{2\sqrt{i}\varphi(z+\xi+H(x))} - 1 & \text{for } z \geq \xi, \end{cases}$$

$$\implies G_{\mathbf{u}}(z; 0) = \frac{i^{3/2} e^{-\sqrt{i}\varphi z}}{\nu_0 \epsilon^2 \varphi \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)} \left[e^{2\sqrt{i}\varphi(z+H(x))} - 1 \right],$$

$$(37) \quad \implies \partial_z G_{\mathbf{u}}(z; \xi)|_{z=-H(x)} = \frac{-e^{-\sqrt{i}\varphi(\xi-H(x))}}{\nu_0 \epsilon^2 \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)} \left(1 + e^{2\sqrt{i}\varphi \xi} \right),$$

$$(38) \quad \implies \partial_z G_{\mathbf{u}}(z; 0)|_{z=-H(x)} = \frac{-2e^{\sqrt{i}\varphi H(x)}}{\nu_0 \epsilon^2 \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)},$$

where $\varphi = \sqrt{(f/\nu_0)}/\epsilon$ (see `Example_barotropic_GeostrophicFlow_v0.2.ipynb`). And then

$$(39) \quad \mathbf{u}(z; x) = \int_{-H(x)}^0 G_u(z; \xi) [\partial_x p(\xi; x) + i\partial_y p(\xi; x)] d\xi - G_u(z; 0) (\tau^x + i\tau^y),$$

The known baroclinic pressure (due to b) adds to the unknown barotropic pressure (due to $p_s(x, y)$) in the integrand of this expression. Otherwise, everything else is known (remember that x in these expressions stands for (x, y) in general).

4. FRICTIONAL GEOSTROPHIC EQUATIONS CLOSURE WITH THE GCVB EQUATION

The problem is closed using the GCVB equation, as follows (*Salmon* 1998, p. 170). This determines the unknown surface pressure field given the surface stress (and vice versa). In particular, work from (17)–(18),

$$(40) \quad \mathfrak{U}(z; x) \equiv U + iV = \int_{-H(x)}^0 \mathbf{u}(z; x) dz,$$

$$(41) \quad = \int_{-H(x)}^0 \left\{ \int_{-H(x)}^0 G_u(z; \xi) [\partial_x p(\xi; x) + i\partial_y p(\xi; x)] d\xi - G_u(z; 0) (\tau^x + i\tau^y) \right\} dz.$$

Computing the z integral of the Green's function $G_u(z; \xi)$ ⁴ gives

$$(42) \quad \begin{aligned} \int_{-H(x)}^0 G_u(z; \xi) dz &= \frac{-ie^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right)}{\nu_0 \epsilon^2 \varphi^2 \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)}, \\ \Rightarrow \int_{-H(x)}^0 G_u(z; 0) dz &= \frac{-i \left(e^{\sqrt{i}\varphi H(x)} - 1 \right)^2}{\nu_0 \epsilon^2 \varphi^2 \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)}. \end{aligned}$$

Therefore,

$$(43) \quad \begin{aligned} \mathfrak{U}(x) &= \frac{-i}{\nu_0 \epsilon^2 \varphi^2 \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)} \times \\ &\quad \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) [\partial_x p(\xi; x) + i\partial_y p(\xi; x)] d\xi \right. \\ &\quad \left. - \left(e^{\sqrt{i}\varphi H(x)} - 1 \right)^2 (\tau^x + i\tau^y) \right\}. \end{aligned}$$

⁴This presumes (correctly) that the denominator in (36) does not vanish.

Also the bottom stress on the fluid is given by

$$\begin{aligned}
 -\partial_z u(z; x)|_{z=-H(x)} &= \int_{-H(x)}^0 \frac{e^{-\sqrt{i}\varphi(\xi-H(x))}}{\nu_0 \epsilon^2 \left(1 + e^{2\sqrt{i}\varphi H(x)}\right)} \left(1 + e^{2\sqrt{i}\varphi \xi}\right) [\partial_x p(\xi; x) + i\partial_y p(\xi; x)] d\xi \\
 &\quad - \frac{2e^{\sqrt{i}\varphi H(x)}}{\nu_0 \epsilon^2 \left(1 + e^{2\sqrt{i}\varphi H(x)}\right)} (\tau^x + i\tau^y), \\
 \implies -\nu_0 \epsilon^2 \partial_z u(z; x)|_{z=-H(x)} &= \int_{-H(x)}^0 \frac{e^{-\sqrt{i}\varphi(\xi-H(x))}}{\left(1 + e^{2\sqrt{i}\varphi H(x)}\right)} \left(1 + e^{2\sqrt{i}\varphi \xi}\right) [\partial_x p(\xi; x) + i\partial_y p(\xi; x)] d\xi \\
 &\quad - \frac{2e^{\sqrt{i}\varphi H(x)}}{\left(1 + e^{2\sqrt{i}\varphi H(x)}\right)} (\tau^x + i\tau^y)
 \end{aligned} \tag{44}$$

using (37) and (38). Notice how the derivatives of the unknown surface pressure part of p end up multiplying expressions that are functions of x .

Eqns. (17) and (18) also imply that:

$$if\mathfrak{U} = -H \left(\frac{\partial p_b}{\partial x} + i \frac{\partial p_b}{\partial y} \right) - \left(\frac{\partial \chi}{\partial y} + i \frac{\partial \chi}{\partial x} \right) + \tau^x + i\tau^y - \tau_b^x - i\tau_b^y. \tag{45}$$

Therefore, for an unstratified fluid ($\chi = 0, \nabla p_b = \nabla p_s$), specifying the surface pressure gradient ∇p_s closes the problem. We compute the surface stress from (45) by substituting (44). Then $u(z), v(z), U, V$ and the force balance can all be computed. See Figure 1 for an example calculation. Alternatively, the surface stress can be specified, and the pressure gradient can be computed.

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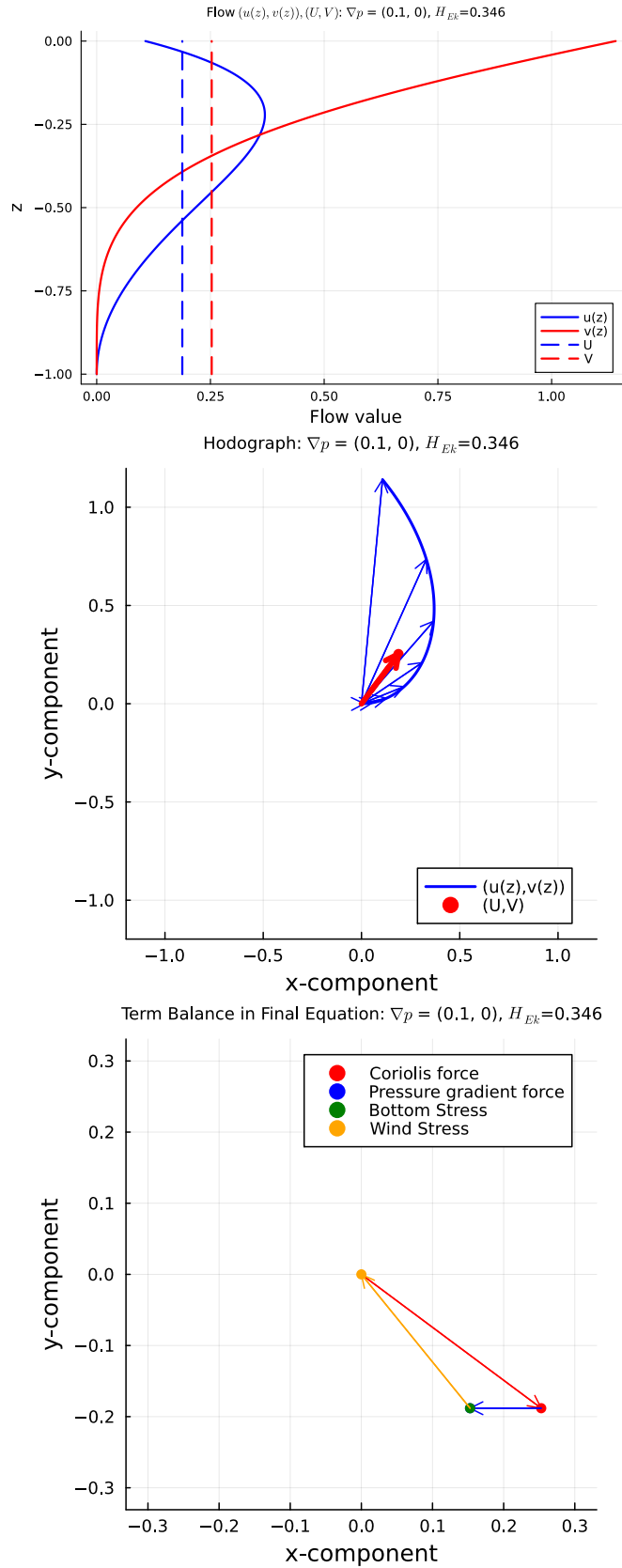


FIGURE 1. Example solution to the barotropic frictional geostrophic flow problem from `Example_barotropic_GeostrophicFlow_v0.2.ipynb`.